

high-quality protein (one that contains all of the essential amino acids) in the hour or two after resistance exercise.^{8,32}

RESEARCH PERSPECTIVE 15.4

The Myth of High-Protein Diets

Trends in dieting for weight loss come and go. For many years, dietary fat received a lot of the blame for increasing rates of overweight and obesity, and fat-free or low-fat diets were popular. However, despite this emphasis on reducing fat intake, rates of overweight and obesity continued to climb. Recently, some people have been making the counterargument that Americans have grown fatter because they consume too many starches and sugars and not enough meat and fat. In 2015, the *Dietary Guidelines Advisory Committee* lifted their recommendation that dietary cholesterol should be restricted, citing evidence that dietary cholesterol does not influence blood cholesterol concentrations all that much. So, is a high-fat, high-protein regimen the new healthy diet du jour?

In an opinion piece published in *The New York Times*, Dean Ornish, a medical doctor and clinical professor of medicine at the University of California San Francisco, makes the argument that animal fats and proteins are not health foods.²⁵ Despite being told for decades to reduce fats and animal proteins in their diets, Americans consumed 67% more added fat and 41% more meat in 2000 than they did in 1950. Additionally, consumption of refined sugars has gone up 39%. Increasing consumption of animal protein increases risk for premature death from all causes, and heavy consumption of saturated fats and trans fats doubles the risk for Alzheimer's disease. Dr. Ornish and his colleagues at the Preventative Research Institute at UCSF have conducted clinical research trials showing the benefits of a plant-based diet for reducing circulating free fatty acids and insulin and reducing inflammation associated with animal protein and fat consumption. Dr. Ornish and others have found that diets rich in fruits, vegetables, whole grains, and polyunsaturated fats and low in saturated fats, refined carbohydrates, and animal proteins actually reverses the progression of cardiovascular diseases and can reduce the need for chronic medications. The more people adhered to the dietary recommendations, the more improvement Dr. Ornish measured, even in the oldest patients. Overall, despite the popularity of high-protein diets, only a whole-foods, plant-based diet has been tested clinically and shown to reverse disease trends.

Protein Intake and Resistance Exercise

In order to increase muscle mass, the rate of protein synthesis (anabolism) must exceed the rate of protein degradation (catabolism). While both processes are important in building muscle mass, for healthy exercisers, increasing the synthesis of new protein plays a much larger role than decreasing degradation. As summarized by Stuart Phillips,²⁶ both resistance training and protein (or amino acid) consumption can independently promote protein synthesis. However, combining the two—that is, consuming protein after a bout of resistance exercise—synergistically stimulates protein synthesis. Based on years of elegant research, we now have better answers to the specific questions of what to eat, how much is necessary, and when to consume proteins in relation to exercise if the goal is an increased muscle mass.

Rapidly digested proteins that contain high proportions of essential amino acids are more effective in stimulating muscle protein synthesis than are other proteins. The key amino acid appears to be leucine, since peak muscle protein synthesis correlates with peak leucine concentration. This finding has led to the concept of leucine as a trigger for promoting protein synthesis.

There is no evidence that commercially available protein supplements are necessary to achieve desired goals. Food protein, particularly high-quality dairy protein—containing all essential amino acids and high in leucine—may be more effective than other protein sources. In particular, high-quality milk-based proteins like casein and whey are effective in promoting gains in muscle mass during periods of intense resistance training.

Consuming protein in close temporal proximity to exercise leads to greater rates of protein synthesis and greater muscle hypertrophy. Current evidence favors consumption of protein after resistance exercise, but the optimal timing after exercise is not known. The anabolic effect of resistance exercise is long lasting, but likely diminishes with increasing time postexercise, so current evidence supports a 1 to 2 h postexercise window for optimal synthesis rates. In addition, athletes should consume protein as part of each meal to promote optimal protein synthesis throughout the day.

Intakes in the range of 1.4 to 1.6 g of protein per kilogram per day should be consumed for the maintenance and building of muscle

mass. This represents a higher overall daily protein intake than the RDA, which is currently 0.8 g of protein/kg body mass per day. Consuming amounts of protein in excess of 2.0 g/kg per day provides no additional benefit. Individual meals should include at least 0.25 g of protein/kg body mass.

In Review

- Carbohydrates are sugars and starches. They exist in the body as monosaccharides, disaccharides, oligosaccharides, and polysaccharides. All carbohydrates must be broken down into monosaccharides before the body can use them as a fuel.
- Insufficient intake of carbohydrate during periods of intense training can lead to depletion of glycogen stores. Conversely, glycogen loading by consumption of a diet rich in carbohydrate offers major benefits to performance.
- Endurance performance can be enhanced when carbohydrate is consumed an hour or more before beginning exercise, within 5 min of starting exercise, and during exercise. Exercisers can replenish carbohydrate stores rapidly by ingesting carbohydrate during the first 2 h of recovery. This can be facilitated by the addition of protein to the carbohydrate supplement.
- Fats, or lipids, exist in the body as triglycerides, FFAs, phospholipids, and sterols. They are stored primarily as triglycerides, which are the body's most concentrated energy source. A triglyceride molecule can be broken down into one glycerol and three fatty acid molecules. Only the FFAs are used by the body for energy production.
- Although fat is a major energy source, the use of high-fat diets to enhance endurance performance by sparing glycogen has generally been unsuccessful.
- The smallest unit of protein is an amino acid. All proteins must be broken down to amino acids before the body can use them. Only nonessential amino acids can be synthesized in the body; the essential amino acids must be attained through diet.
- Protein is not a primary energy source in our bodies, but it can be used for energy production during endurance exercise.
- The current RDA for protein (0.8 g/kg per day) may be too low for athletes involved in intense resistance training or for endurance athletes. In most cases, athletes should consume 1.2 to 1.7 g/kg per day. During the initial days of training, during periods of very intense training, or when insufficient energy is consumed, the requirement might be higher. However, extremely high-protein diets offer no

additional benefits and could pose a health risk to normal kidney function and contribute to risk of dehydration.

- Protein supplementation during recovery from resistance training can stimulate muscle protein synthesis.
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Vitamins

Vitamins are a group of organic compounds that perform specific functions in the body. We need them in relatively small quantities to fully use the other nutrients we ingest. Vitamins act primarily as catalysts or cofactors in chemical reactions. They are essential for energy release, tissue building, and metabolic regulation. Vitamins can be classified into one of two major categories: fat soluble and water soluble. The fat-soluble vitamins, A, D, E, and K, are absorbed from the digestive tract bound to lipids (fats). These vitamins are stored in the body, so excessive intake can cause toxic accumulations. Vitamin C and the B-complex vitamins biotin, pantothenic acid, and folate are water soluble. They are absorbed from the digestive tract along with water. Any excess of these vitamins is excreted, mostly in the urine, but vitamin toxicity has been reported with high doses of some of these. [Table 15.3](#) lists the various vitamins and their RDA values, or Adequate Intake (AI) values when the RDA values are not available.

Most vitamins have some functions important to the athlete:

- Vitamin A is crucial for normal growth and development because it plays an integral role in bone development.
- Vitamin D is essential for intestinal absorption of calcium and phosphorus and thus for bone development and strength. By regulating calcium absorption, this vitamin also has a key role in neuromuscular function.
- Vitamin K is an intermediate in the electron transport chain, making it important for oxidative phosphorylation.

Of all the vitamins, the B-complex vitamins and vitamins C and E have been most extensively studied for their potential to facilitate athletic performance. In the following sections, we briefly consider these vitamins.

B-Complex Vitamins

The B-complex vitamins were once thought to be a single vitamin. Now more than a dozen B-complex vitamins have been identified. These vitamins play essential roles in cellular metabolism, primarily by serving as cofactors in various enzyme systems involved in the oxidation of food and the production of energy. Vitamin B₁ (thiamin) is needed for the conversion of pyruvic acid to acetyl coenzyme A. Vitamin B₂ (riboflavin) becomes flavin adenine dinucleotide (FAD), which acts as a hydrogen acceptor during oxidation. Vitamin B₃ (niacin) is a component of nicotinamide adenine dinucleotide phosphate (NADP), a coenzyme in glycolysis. Vitamin B₁₂ has a role in amino acid metabolism and is also needed for the production of red blood cells, which transport oxygen to the cells for oxidation. The B-complex vitamins have such a close interrelationship that a deficiency in one can impair utilization of the others. Symptoms of deficiencies vary with the vitamins involved.

A few studies have shown that supplementation of one or more of the B-complex vitamins facilitates performance. However, most researchers agree that this is true only if the individual being studied has a preexisting B-complex deficiency. Creating a deficiency in one or more of the B-complex vitamins usually impairs performance, but this is reversed when the deficiency is corrected with supplementation. No compelling evidence supports supplementation when there is no deficiency.

Vitamin C

Vitamin C (ascorbic acid) is common in our foods, but deficiencies can occur in people who smoke, use oral contraceptives, have surgery, or run a fever. This vitamin is important for the formation and maintenance of collagen, a crucial protein found in connective tissue, so it is essential for healthy bones, ligaments, and blood vessels. Vitamin C also functions in

- the metabolism of amino acids;
- synthesis of some hormones, such as the catecholamines (epinephrine and norepinephrine) and the anti-inflammatory corticoids; and
- promotion of iron absorption from the intestines.

Some people also believe that vitamin C assists healing, combats fever and infection, and prevents or cures the common cold, but evidence of these attributes is inconclusive.

Vitamin C supplementation to enhance performance has produced equivocal findings in the research conducted to date. As with other vitamins, vitamin C supplementation does not improve performance when no deficiency exists.

Vitamin E

Vitamin E is stored in muscle and fat. This vitamin's functions are not well established, although it is known to enhance the activity of vitamins A and C by preventing their oxidation. Indeed, the most important role of vitamin E is its action as an antioxidant. It disarms **free radicals** (highly reactive molecules) that could otherwise severely damage cells and disrupt metabolic processes. Exercise has been shown to cause DNA damage to the cell, whereas supplementing the intake of vitamin E reduces DNA damage induced by exercise. However, investigators found no benefit of 30 days of vitamin E supplementation with respect to the muscle damage that resulted from 240 maximal isokinetic eccentric knee flexion–extension actions (24 sets of 10 repetitions each) when compared with a placebo control condition.⁷

TABLE 15.3 RDAs or AIs for Vitamins and Minerals

		9-13 years		14-18 years		19-50 years		51-70 years	
Vitamin or Mineral	Dose	Male	Female	Male	Female	Male	Female	Male	Female
Vitamins									
A (retinol)	µg/day	600	600	900	700	900	700	900	700
B ₁ (thiamine)	mg/day	0.09	0.09	1.2	1.0	1.2	1.1	1.2	1.2
B ₂ (riboflavin)	mg/day	0.9	0.9	1.3	1.0	1.3	1.1	1.3	1.1
B ₃ (niacin)	mg/day	12	12	16	14	16	14	16	14
B ₆	mg/day	1.0	1.0	1.3	1.2	1.3	1.3	1.7	1.5
B ₁₂	µg/day	1.8	1.8	2.4	2.4	2.4	2.4	2.4	2.4
C	mg/day	45	45	75	65	90	75	90	75
D	µg/day	15 ^a	15 ^a	15 ^a	15 ^a	15 ^a	15 ^a	15 ^a	15 ^a
E	mg/day	11	11	15	15	15	15	15	15
Biotin (vitamin H)	µg/day	20 ^a	20 ^a	25 ^a	25 ^a	30 ^a	30 ^a	30 ^a	30 ^a
K	µg/day	60 ^a	60 ^a	75 ^a	75 ^a	120 ^a	90 ^a	120 ^a	90 ^a
Folate	µg/day	300	300	400	400	400	400	400	400
Pantothenic acid	mg/day	4 ^a	4 ^a	5 ^a	5 ^a	5 ^a	5 ^a	5 ^a	5 ^a
Minerals									
Calcium	mg/day	1,300 ^a	1,300 ^a	1,300 ^a	1,300 ^a	1,000 ^a	1,000 ^a	1,000 ^a	1,200 ^a
Chloride	g/day	2.3 ^a	2.3 ^a	2.3 ^a	2.3 ^a	2.3 ^a	2.3 ^a	2.0 ^a	2.0 ^a
Chromium	µg/day	25 ^a	21 ^a	35 ^a	24 ^a	35 ^a	25 ^a	30 ^a	20 ^a
Copper	µg/day	700	700	890	890	900	900	900	900
Fluoride	mg/day	2 ^a	2 ^a	3 ^a	3 ^a	4 ^a	3 ^a	4 ^a	3 ^a
Iodine	µg/day	120	120	150	150	150	150	150	150
Iron	mg/day	8	8	11	15	8	18	8	8
Magnesium	mg/day	240	240	410	360	410 ^b	315 ^b	420	320
Manganese	mg/day	1.9 ^a	1.6 ^a	2.2 ^a	1.6 ^a	2.3 ^a	1.8 ^a	2.3 ^a	1.8 ^a
Molybdenum	µg/day	34	34	43	43	45	45	45	45
Phosphorus	mg/day	1,250	1,250	1,250	1,250	700	700	700	700
Potassium	g/day	4.5 ^a	4.5 ^a	4.7 ^a	4.7 ^a	4.7 ^a	4.7 ^a	4.7 ^a	4.7 ^a
Selenium	µg/day	40	40	55	55	55	55	55	55
Sodium	g/day	1.5 ^a	1.5 ^a	1.5 ^a	1.5 ^a	1.5 ^a	1.5 ^a	1.3 ^a	1.3 ^a
Zinc	mg/day	8	8	11	9	11	8	11	8

^aAdequate Intake (AI). RDA is not available.

^bMen: age 19 to 30 years = 400; age 31 to 50 years = 420. Women: age 19 to 30 years = 310; age 31 to 50 years = 320.

Note. Values are also available for infants and small children and for women during pregnancy and lactation.

From USDA National Agricultural Library. Available: <http://fnic.nal.usda.gov/dietary-guidance/dietary-reference-intakes/dri-tables>

Many athletes have consumed supplementary doses of vitamin E since it has been postulated to benefit performance through its relationship with oxygen use and energy supply. However, research reviews generally conclude that vitamin E supplementation does not improve athletic performance.

Minerals

A number of inorganic substances known as minerals are essential for normal cellular functions. Minerals account for approximately 4% of body weight. Some are present in high concentrations in the skeleton and teeth, but minerals are also found throughout the body, in and around every cell, dissolved in the body's fluids. They can be present either as ions or combined with various organic compounds. Mineral compounds that can dissociate into ions in the body are called **electrolytes**.

By definition, **macrominerals** are those minerals needed by the body in amounts greater than 100 mg per day. **Microminerals**, or **trace elements**, are those needed in smaller amounts. [Table 15.3](#) lists the essential minerals and their RDAs or AIs.

Mineral intake is less likely to be supplemented by athletes than vitamin intake, possibly because far fewer claims have been made about the performance-enhancing qualities of specific minerals. Of the minerals, calcium and iron have been most frequently investigated.

Calcium

Calcium is the most abundant mineral in the body, constituting approximately 40% of the total mineral content. Calcium is well known for its importance in building and maintaining healthy bones, and that is where most of it is stored. But it is also essential for nerve impulse transmission. Calcium plays major roles in enzyme activation and regulation of cell membrane permeability, both important for metabolism. This mineral is also essential for normal muscle function: Recall from [chapter 1](#) that calcium is stored in the sarcoplasmic reticulum of muscles and released when the muscle fibers are stimulated. It is required for formation of the actin–myosin cross-bridges that cause the fibers to contract.

Sufficient calcium intake is critical to our health. If we do not consume enough calcium, it is removed from its storage sites in the body, especially the bones. This condition is called osteopenia. It weakens the bones and can lead to osteoporosis, a common problem in postmenopausal women and aging men and women. Unfortunately, few studies have been conducted on calcium supplementation, but their results suggest that supplementation is of

no value in the presence of an adequate (RDA) dietary intake of calcium.

Phosphorus

Phosphorus is closely linked to calcium. It constitutes approximately 22% of the body's total mineral content. About 80% of this phosphorus is combined with calcium (calcium phosphate), providing strength and rigidity to the bones. Phosphorus is an essential part of metabolism, cell membrane structure, and the buffering system to maintain constant blood pH. Phosphorus plays a major role in bioenergetics: It is an essential component of adenosine triphosphate. There is no evidence to suggest the need for supplementation in athletes.

Iron

Iron—a micromineral—is present in the body in relatively small amounts (35-50 mg/kg of body weight). It plays a crucial role in oxygen transportation: Iron is required for the formation of both hemoglobin and myoglobin. Hemoglobin, located in the red blood cells, binds with oxygen in the lungs and then transports it to the body tissues via the blood. Myoglobin, found in muscle, combines with oxygen and stores it until needed.

Iron deficiency is prevalent throughout the world. By some estimates, as much as 25% of the world's population is iron deficient. In the United States, approximately 20% of women and 3% of men are iron deficient, as are 50% of pregnant women. The major problem associated with this condition is iron-deficiency anemia, in which hemoglobin levels are reduced, decreasing the blood's oxygen-carrying capacity. This causes fatigue, headaches, and other symptoms. Iron deficiency is a more common problem in women than in men because both menstruation and pregnancy cause iron losses that must be replenished. This problem is compounded by the fact that women generally consume less food, and thus less iron, than men.

Iron has received much attention in the research literature. Women are considered anemic when their hemoglobin concentration is below 10 g per 100 ml of blood, and men are considered to be anemic below values of 12 g per 100 ml of blood. Studies suggest that 22%

to 25% of female athletes and 10% of male athletes are iron deficient, but these numbers may be conservative. Hemoglobin concentration is not the only marker of anemia or necessarily the best. Plasma ferritin concentration provides a good marker of the body's iron stores. Values below 20 to 30 µg/L indicate low stores of body iron.

When iron supplements are given to those who are iron deficient (i.e., with low plasma ferritin levels), endurance performance typically improves. However, supplementation of iron in those who are not deficient has little or no benefit. In fact, iron supplements can be a health risk, because excess iron is toxic for the liver, and ferritin levels higher than 200 µg/L are associated with an increased risk for coronary artery disease.

Sodium, Potassium, and Chloride

Sodium, potassium, and chloride are the important electrolytes that are distributed throughout all body fluids and tissues. Sodium and chloride are found primarily in the fluid outside of the cells and in the blood plasma, but potassium is located mainly inside the cells. This selective distribution of these three minerals establishes the separation of electrical charge across neuron and muscle cell membranes. Thus, these minerals enable neural impulses to control muscle activity (see [chapter 3](#)). In addition, they are responsible for maintaining the body's water balance and distribution, normal osmotic equilibrium, acid–base balance (pH), and normal cardiac rhythm ([chapter 6](#)).

Western diets are high in sodium, so sodium deficiency is an unlikely occurrence. However, minerals are lost with sweating, so excessive sweating during extreme exertion or exercise in a hot environment can deplete these minerals. When discussing mineral imbalances, we often focus on deficiencies. However, many of these minerals also have negative effects when taken in excess. In fact, excess potassium can cause heart failure! Individual needs vary, but megadoses are never advisable.

To conclude this section on vitamins and minerals, we can say that while physical activity increases vitamin and mineral requirements, the need is generally met by adequate food intake. For athletes who eat balanced meals that meet their caloric needs, it is highly likely that all vitamin and mineral needs will be met and supplementation

will have no performance benefits. However, for those who are intentionally consuming a low-energy or unbalanced diet, supplementation may be necessary to maintain health and performance. If there is any question about the adequacy of an athlete's diet, a low-dose multivitamin and mineral supplement may be appropriate.

The previous discussion on nutrient intakes provides several RDAs. In the early 1940s, the Food and Nutrition Board of the National Academy of Sciences established the United States Recommended Daily Allowance (RDA) guidelines for all food nutrients. The latest edition of the RDAs in their original form was released in 1989. The RDAs provide estimates of safe and adequate daily dietary intakes and estimated minimum requirements for selected vitamins and minerals. A major revision of the RDAs was initiated in the early 1990s. The RDAs have been replaced by new recommendations called Dietary Reference Intakes (DRIs). The DRIs reflect a joint effort between the United States and Canada to provide dietary intake recommendations grouped by nutrient function and classification.

The new DRIs were released in a series of reports starting in 1997 and continuing through 2005. They include four different reference values:

- Estimated Average Requirement (EAR)—the intake value estimated to meet the requirement for 50% of healthy individuals in an age- and sex-specific group
- Recommended Dietary Allowance (RDA)—the intake value sufficient to meet the nutrient requirements of nearly all (97%-98%) individuals in a specific group
- Tolerable Upper Intake Level (UL)—the highest level of daily nutrient intake that is unlikely to pose no risk of adverse health effects to almost all individuals in a specific group
- Adequate Intake (AI)—a recommended intake value based on observed or experimentally determined approximations or estimates of nutrient intake by healthy individuals in a specific group that are assumed to be adequate; this is used when an RDA cannot be determined

For further information on DRIs and specific recommendations for each of the nutrient classifications by sex and age, refer to the U.S. Department of Agriculture's Food and Nutrition Information Center at www.nal.usda.gov/fnic.

In Review

- Vitamins perform numerous functions in the body and are essential for normal growth and development. Many are involved in metabolic processes, such as those leading to energy production.
- Vitamins A, D, E, and K are fat soluble. These can accumulate to toxic levels in the body. Vitamin C and the B-complex vitamins biotin, pantothenic acid, and folate are water soluble. Excesses of these are excreted, so toxicity is rarely a problem. Several of the B-complex vitamins are involved in the processes of energy production.
- Macrominerals are minerals required in amounts of more than 100 mg per day. Microminerals (trace elements) are those we require in smaller amounts.
- Minerals are required for numerous physiological processes, such as muscle contraction, oxygen transport, fluid balance, and bioenergetics. Minerals can dissociate into ions, which can participate in numerous chemical reactions. Minerals that can dissociate into ions are called electrolytes.
- Vitamins and minerals do not appear to have any special performance-enhancing value. Taking them in amounts greater than the RDA will not improve performance and may have unwanted effects.

Water

Water is often not thought of as a nutrient because it has no caloric value. Yet its importance in maintaining life is perhaps second only to that of oxygen. Water constitutes about 60% of a typical young man's and 50% of a typical young woman's total body weight, but this varies with body composition, since the fat-free mass has a much higher water content (~73% water) than the fat mass (~10% water). It has been estimated that we can survive losses of up to 40% of our body weight in fat, carbohydrate, and protein. But a water loss of only 9% to 12% of body weight can be fatal.

Approximately two-thirds of the water in our bodies is contained in our cells and is referred to as **intracellular fluid**. The remainder is outside the cells, referred to as the **extracellular fluid**. Extracellular

fluid includes the interstitial fluid surrounding the cells, the blood plasma, lymph, and other body fluids.

Water plays several critical roles in exercise. Among its most important functions, water provides transportation between and delivery to the body's various tissues, regulates body temperature, and maintains blood pressure for proper cardiovascular function. In the next sections, we more closely examine the role of water in exercise and performance.

Water and Electrolyte Balance

For optimal performance, the body's water and electrolyte contents should remain relatively constant. Unfortunately, this doesn't always happen during exercise. In the next sections, we examine water content and electrolyte balance at rest, how exercise affects these, and the impact on performance when water or electrolyte balance is disturbed.

Water Balance at Rest

Under normal resting conditions, the body's water content is relatively constant: Water intake equals water output. About 60% of our daily water intake is obtained from the fluids we drink and about 30% is from the foods we consume. The remaining 10% is produced in our cells during metabolism (recall from [chapter 2](#) that water is a by-product of oxidative phosphorylation). Metabolic water production varies from 150 to 250 ml per day, depending on the rate of energy expenditure: Higher metabolic rates produce more water. The total daily water intake from all sources averages about 33 ml per kilogram of body weight per day. For a 70 kg (154 lb) person, average intake is 2.3 L per day. Water output, or water loss, occurs from four sources:

- Evaporation from the skin
- Evaporation from the respiratory tract
- Excretion from the kidneys
- Excretion from the large intestine

Human skin is permeable to water. Water diffuses to the skin's surface, where it evaporates into the environment. In addition, the gases we breathe are constantly being humidified by water as they

pass through the respiratory tract. These two types of water loss (from the skin and respiration) occur without our sensing them. Thus, they are termed insensible water losses. Under cool, resting conditions, these losses account for about 30% of daily water loss.

The majority of our daily water loss—60% at rest—occurs from our kidneys, which excrete water and waste products as urine. Under resting conditions, the kidneys excrete about 50 to 60 ml of water per hour. Another 5% of the water is lost by sweating (although this is often considered along with insensible water loss), and the remaining 5% is excreted from the large intestine in the feces. The sources of water gain and water loss at rest are depicted in [figure 15.13](#).

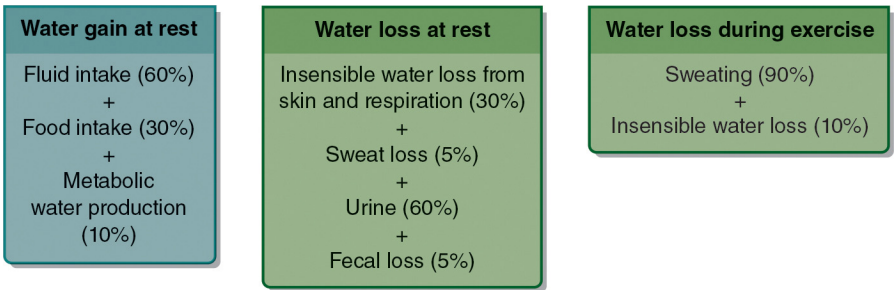


FIGURE 15.13 Sources and approximate percentages of body water gains and losses at rest and during exercise (see text for more detail).

Water Balance During Exercise

Water loss accelerates during exercise, as seen in [table 15.4](#). The ability to lose the heat generated during exercise depends primarily on the formation and evaporation of sweat. As body temperature increases, sweating increases in an effort to prevent overheating (see [chapter 12](#)). But at the same time, more water is produced during exercise because of increased oxidative metabolism. Unfortunately, the amount produced even during the most intense effort has only a small impact on the **dehydration**, or water loss, that results from heavy sweating.

TABLE 15.4 Typical Values of Water Loss From the Body at Rest in a Cool Environment and During Prolonged Exhaustive Exercise

Source of loss	Resting		Prolonged exercise	
	ml/h	% total	ml/h	% total

Skin (insensible loss)	15	15	15	1
Respiration (insensible loss)	15	15	100	7
Sweating	4	5	1,200	91
Urine	58	60	10	1
Feces	4	5		0
Total	96	100	1,325	100

In general, the volume of sweat produced during exercise is determined by metabolic rate, environmental temperature, radiant heat load, humidity, air velocity, clothing, and body size.

These factors influence the body's heat storage and temperature regulation. Heat is transferred from warmer areas to cooler ones, so heat loss from the body is impaired by high environmental temperatures, radiation, high humidity, and still air. Body size, specifically the ratio between surface area and mass, is important because large individuals generally expend more energy to do a given task, so they typically have higher metabolic rates and produce more heat. But they also have more surface area (skin), which allows more sweat formation and evaporation. As exercise intensity increases, so does the metabolic rate. This increases body heat production, which in turn increases sweating. To conserve water during exercise, blood flow to the kidneys decreases in an attempt to prevent dehydration; but like the increase in metabolic water production, this may be insufficient. Athletes commonly lose between 1% and 6% of their body water during intense, prolonged exercise. During high-intensity exercise under environmental heat stress, sweating can cause losses of as much as 3 L of water per hour. ([Chapter 12](#) contains additional information about body water losses during exercise in warm environments.) In cold, dry environments or at altitude, water loss from respiration contributes to the overall loss of body water as well.

Dehydration and Exercise Performance

Even minimal changes in the body's water content can impair endurance performance. Without adequate fluid replacement, an athlete's exercise tolerance shows a pronounced decrease during long-term activity because of water loss through sweating. The impact of dehydration on the cardiovascular and thermoregulatory systems is quite predictable. Fluid loss decreases plasma volume. This decreases blood pressure, which in turn reduces blood flow to the muscles and skin. In an effort to overcome this, heart rate

increases. Because less blood reaches the skin, heat dissipation is hindered, and the body retains more heat. Thus, when a person is dehydrated by 2% of body weight or more, both heart rate and body temperature are elevated during exercise above values observed when normally hydrated.

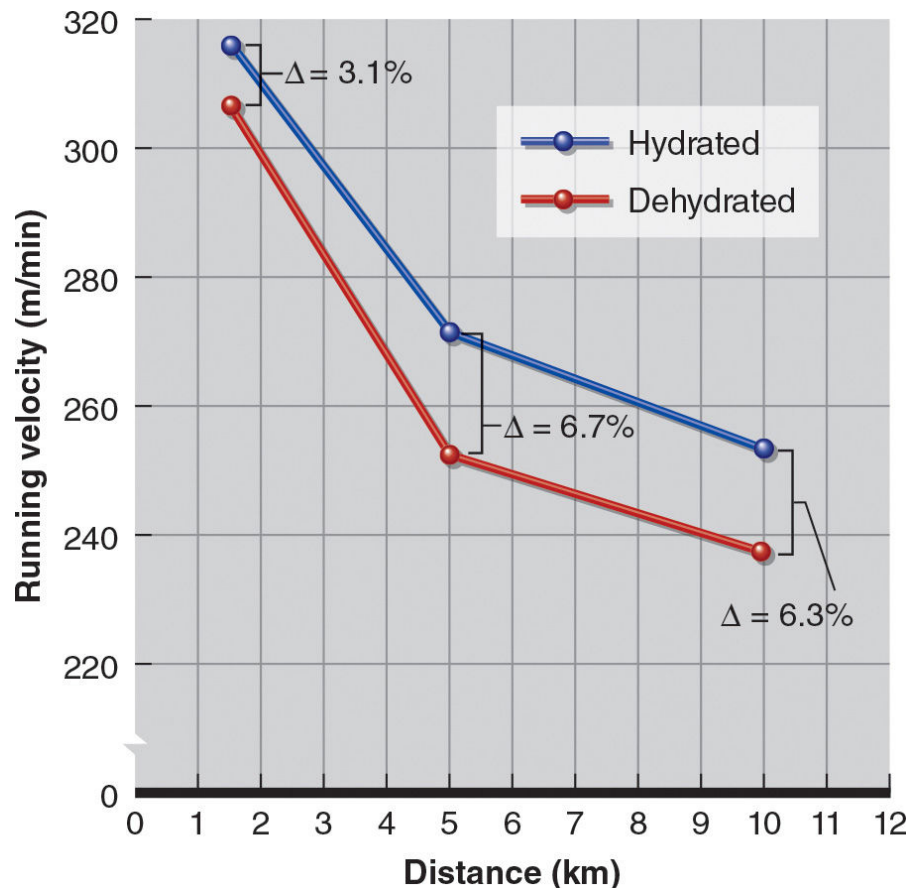


FIGURE 15.14 The decline in running velocity (meters per minute) with dehydration of about 2% of body weight for 1,500 m, 5,000 m, and 10,000 m time trials compared with velocity in the normally hydrated condition.

Reprinted by permission from L.E. Armstrong, D.L. Costill, and W.J. Fink, "Influence of Diuretic-Induced Dehydration on Competitive Running Performance," *Medicine and Science in Sports and Exercise* 17 (1985): 456-461.

As one might expect, these physiological changes decrease exercise performance. Figure 15.14 illustrates the effects of an approximate 2% decrease in body weight attributable to dehydration from the use of a diuretic on distance runners' performance in 1,500 m, 5,000 m, and 10,000 m time trials on an outdoor track.³ The dehydration condition resulted in plasma volume decreases between 10% and 12%. Although the average $\dot{V}O_{2\max}$ did not differ between

the normally hydrated and dehydrated trials, mean running velocity decreased by 3% in the 1,500 m run and by more than 6% in the 5,000 and 10,000 m runs. The greater the duration of the performance, the greater is the expected decline in performance for the same degree of dehydration. These trials were conducted in relatively cool weather. The higher the temperature, humidity, and radiation, the greater the expected decrement in performance for the same degree of dehydration. The decrement in performance would be progressively greater with greater levels of dehydration.

The effect of dehydration on performance in muscular strength, muscular endurance, and anaerobic types of activities is not as clear. Decrements have been seen in some studies, whereas other studies have shown no change in performance. In one of the best-controlled studies, researchers at Penn State University reported that 2% dehydration resulted in significant deterioration of basketball skills in 12- to 15-year-old boys who were skilled basketball players.¹⁷

Wrestlers and other weight-category athletes commonly dehydrate to get a weight advantage during the weigh-in for a competition. Most rehydrate after the weigh-in before the competition and experience only small decrements in performance. A summary of the effects of dehydration on exercise performance is shown in [table 15.5](#).

In Review

- Water balance depends on electrolyte balance, and vice versa.
- At rest, water intake equals water output. Water intake includes water ingested from foods and fluids and produced as a metabolic by-product. The majority of water output at rest occurs from the kidneys, but water also is lost from the skin, from the respiratory tract, and in feces.
- During exercise, metabolic water production increases as metabolic rate increases.
- Water loss during exercise increases because as heat in the body increases, more water is lost with increased sweating. Sweat becomes the primary avenue for water loss during exercise. In fact, the kidneys decrease urine production in an effort to prevent dehydration.
- When dehydration reaches 2% of body weight, aerobic endurance performance—and even skilled performance in sports such as shooting free throws in basketball

—is notably impaired. Heart rate and body temperature increase in response to dehydration.

Electrolyte Balance During Exercise

Normal body function depends on a balance between water and electrolytes. We have discussed the effects of water loss on performance. Now we turn our attention to the effects of the other component of this delicate balance: electrolytes. When large amounts of water are lost from the body, as during exercise, the balance between water and electrolytes can be disrupted quickly. In the next sections, we examine the effects of exercise on electrolyte balance. Our focus is on the two major routes for electrolyte loss: sweat and urine production.

Electrolyte Loss in Sweat Human sweat is a filtrate of blood plasma, so it contains many substances found there, including sodium (Na^+), chloride (Cl^-), potassium (K^+), magnesium (Mg^{2+}), and calcium (Ca^{2+}). Although sweat tastes salty, it contains far fewer minerals than the plasma and other body fluids. In fact, sweat is 99% water.

Sodium and chloride are the predominant ions in sweat and blood. As indicated in [table 15.6](#), the concentrations of sodium and chloride in sweat are about one-third those found in plasma and five times those found in muscle. Each of these three fluids' **osmolarity**, which is the ratio of solutes (such as electrolytes) to fluid, is also shown. Sweat's electrolyte concentration can vary considerably between individuals. It is strongly influenced by genetics, the rate of sweating, the state of training, and the state of heat acclimatization.

At the elevated rates of sweating reported during endurance events, sweat contains large amounts of sodium and chloride but little potassium, calcium, and magnesium. Based on estimates of the athlete's total body electrolyte content, such losses would lower the body's sodium and chloride content by only about 5% to 7%. Total body levels of potassium and magnesium, two ions principally confined to the insides of cells, would decrease by about 1%. These losses probably have no measurable effect on an athlete's performance.

As electrolytes are lost in sweat, the remaining ions are redistributed among the body tissues. Consider potassium. It diffuses

from active muscle fibers as they contract, entering the extracellular fluid. The increase this causes in extracellular potassium levels does not equal the amount of potassium that is released from active muscles, because potassium is taken up by inactive muscles and other tissues while the active muscles are losing it. During recovery, intracellular potassium levels normalize quickly. Some researchers suggest that these muscle potassium disturbances during exercise might contribute to fatigue by altering the membrane potentials of neurons and muscle fibers, making it more difficult to transmit impulses.

TABLE 15.5 Alterations in Physiological Function and Performance During Dehydration of 2% or Greater

Variable	Dehydration
Cardiovascular	
Blood volume/Plasma volume	Decreased
Cardiac output	Decreased
Stroke volume	Decreased
Heart rate	Increased
Metabolic	
Aerobic capacity ($\dot{V}O_{2\max}$)	No change or decreased
Anaerobic power (Wingate test)	No change or decreased
Anaerobic capacity (Wingate test)	No change or decreased
Blood lactate, peak value	Decreased
Buffer capacity of the blood	Decreased
Lactate threshold, velocity	Decreased
Muscle and liver glycogen	Decreased
Blood glucose during exercise	Possibly decreased
Protein degradation with exercise	Possibly decreased
Thermoregulation and fluid balance	
Electrolytes, muscle and blood	Decreased
Exercise core temperature	Increased
Sweat rate	Decreased, delayed onset
Skin blood flow	Decreased
Performance	
Muscular strength	No change or decreased
Muscle endurance	No change or decreased
Muscular power	Unknown
Speed of movement	No change or decreased
Run time to exhaustion	Decreased
Total work performed	Decreased
Attention and focus	Decreased
Some skill aspects of performance	Decreased

Note. Data for this table were derived from the following reviews: M. Fogelholm, 1994, "Effects of bodyweight reduction on sports performance," *Sports Medicine* 18: 249-267; C.A. Horswill, 1994, "Physiology and nutrition for wrestling," in D.R. Lamb, H.G. Knutten, & R. Murray (Eds.), *Physiology and nutrition for competitive sport* (Vol. 7, pp. 131-174); H.L. Keller, S.E. Tolly, & P.S. Freedson, 1994, "Weight loss in adolescent wrestlers," *Pediatric Exercise Science* 6: 211-224; and R. Opplinger, H. Case, C. Horswill, G. Landry, & A. Shelter, 1996, "Weight loss in wrestlers: An American College of Sports Medicine position stand," *Medicine and Science in Sports and Exercise* 28: ix-xii.

TABLE 15.6 Electrolyte Concentrations and Osmolarity in Sweat, Plasma, and Muscle of Men Following 2 h of Exercise in the Heatw

Site	Electrolytes (mEq/L)				Osmolarity (mOsm/L)
	Na ⁺	Cl ⁻	K ⁺	Mg ²⁺	
Sweat	40-60	30-50	4-6	1.5-5	80-185
Plasma	140	101	4	1.5	295
Muscle	9	6	162	31	295

Note. mEq/L = milliequivalents per liter (thousandths of 1 g of solute per 1 L of solvent).

Electrolyte Loss in Urine In addition to clearing wastes from the blood and regulating water levels, the kidneys also regulate the body's electrolyte content. Urine production is the other major source of electrolyte loss. At rest, electrolytes are excreted in the urine as necessary to maintain homeostatic levels, and this is the primary route for electrolyte loss. But as the body's water loss increases during exercise, urine production rate decreases considerably in an effort to conserve water. Consequently, with very little urine being produced, electrolyte loss by this avenue is minimized.

The kidneys play another role in electrolyte management. If, for example, a person ingests 250 mEq of salt (NaCl), the kidneys will normally excrete 250 mEq of these electrolytes to keep the body's NaCl content constant. Heavy sweating and dehydration, however, trigger the release of the hormone aldosterone from the adrenal gland. This hormone stimulates renal reabsorption of sodium. Consequently, the body retains more sodium than usual during the hours and days after a prolonged exercise bout. This elevates the body's sodium content and increases the osmolality of the extracellular fluids.

This increased sodium content triggers thirst, compelling the person to consume more water, which is then retained in the extracellular compartment. The increased water consumption reestablishes normal osmolality in the extracellular fluids but leaves these fluids expanded, which dilutes the other substances present there. This expansion of the extracellular fluids has no negative effects and is temporary. In fact, this is one of the major mechanisms for the increase in plasma volume that occurs with training and with acclimatization to exercise in the heat. Fluid levels return to normal within 48 to 72 h after exercise, provided there are no subsequent exercise bouts.

In Review

- The loss of large volumes of sweat can disrupt electrolyte balance, although the concentrations of electrolytes in sweat are much lower than in plasma.
- Electrolyte loss during exercise occurs primarily with water loss from sweating. Sodium and chloride are the most abundant electrolytes in sweat.
- There is substantial variability among individuals in both sweating rate and the electrolyte composition of the sweat. This makes it virtually impossible to create a “one size fits all” plan for replacing fluids and electrolytes.
- At rest, excessive electrolytes are excreted in the urine by the kidneys. But urine production declines substantially during exercise, so little electrolyte loss occurs by this route.
- Dehydration induces the hormones ADH (antidiuretic hormone) and aldosterone to promote renal retention of water and sodium. The increased sodium and decreased fluid volume trigger thirst (see [chapter 4](#)).

Replacement of Body Fluid Losses

The body loses more water than electrolytes during heavy sweating. This raises the osmotic pressure in the body fluids because the electrolytes become more concentrated. The need to replace body water is greater than the need for electrolytes because only by replenishing water content can the electrolytes return to normal concentrations. But how does the body know when this is necessary?

Thirst When people feel thirsty, they drink. The thirst sensation is regulated largely by the osmoreceptors in the hypothalamus. Sensory signals of thirst are invoked when the plasma's osmolality is increased above a threshold value. A second set of signals arises from the low-pressure baroreceptors when low blood volume is sensed. However, relative to osmolar control of thirst, a large blood volume loss is necessary for this backup control system to be activated. Unfortunately, the body's **thirst mechanism** doesn't precisely gauge its state of dehydration. It does not sense thirst until well after dehydration begins. Even when dehydrated, people might desire fluids only at intermittent intervals.

The control of thirst is not fully understood. When permitted to drink water as their thirst dictates, people can require 24 to 48 h to completely replace water lost through heavy sweating. Because of

our sluggish drive to replace body water and to prevent chronic dehydration, we are advised to drink more fluid than our thirst dictates. Because of the increased water loss during exercise, it is imperative that athletes' water intake be sufficient to meet their bodies' needs, and it is essential that they rehydrate during and after an exercise bout.

Benefits of Fluids During Exercise Drinking fluids during prolonged exercise, especially in hot weather, has obvious benefits. Water intake minimizes dehydration, increases in body temperature, cardiovascular stress, and declines in performance. As seen in [figure 15.15](#), when subjects became dehydrated during several hours of treadmill running in the heat (40 °C, or 104 °F) without fluid replacement, their heart rates increased steadily throughout the exercise.⁵ When they were deprived of fluids, the subjects became exhausted and couldn't complete the 6 h exercise. Ingesting either water or a saline solution in amounts equal to weight loss prevented dehydration and kept subjects' heart rates lower. Even warm fluids (near body temperature) provide some protection against overheating, but cold fluids enhance body cooling because some of the deep body heat is used to warm cold drinks to body temperature.

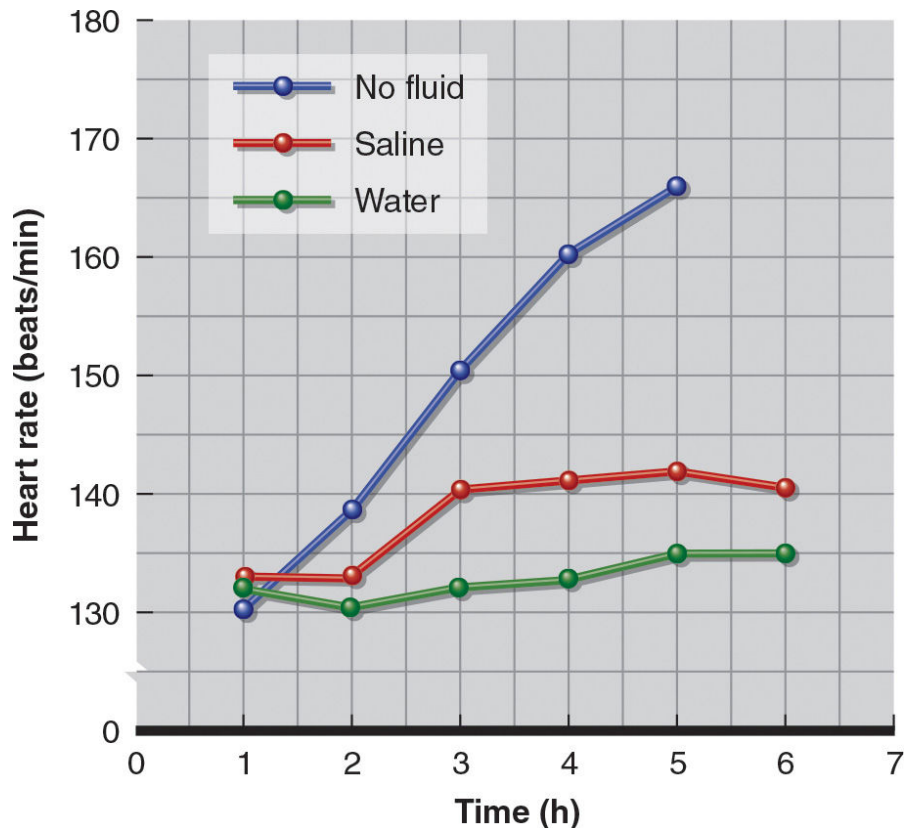


FIGURE 15.15 Heart rate during 6 h of treadmill running in the heat. Subjects received no fluid, a saline solution, or water. Subjects deprived of fluids became exhausted and could not complete the 6 h exercise.

Data from Barr, Costill, and Fink (1991).



Hyponatremia

Fluid replacement is beneficial, but too much of a good thing is problematic. In the 1980s, the first cases of hyponatremia were reported in endurance athletes. **Hyponatremia** is clinically defined as a serum sodium concentration below the normal range of 135 to 145 mmol/L. Symptoms of hyponatremia generally appear when serum sodium levels drop below 130 mmol/L. Early signs and symptoms include bloating, puffiness, nausea, vomiting, and headache. As the severity increases, due to increasing cerebral edema (swelling of the brain), the symptoms include confusion, disorientation, agitation,

seizures, pulmonary edema, coma, and death. How likely is hyponatremia to occur?

The processes that regulate fluid volumes and electrolyte concentrations are highly effective, so consuming enough water to dilute plasma electrolytes is difficult under normal circumstances. Marathoners who lose 3 to 5 L of sweat and drink 2 to 3 L of water maintain normal plasma concentrations of sodium, chloride, and potassium. And distance runners who run 25 to 40 km (15.5-24.9 mi) per day in warm weather and do not salt their food don't develop electrolyte deficiencies.

Some research has suggested that during ultramarathon running (more than 42 km, or 26.2 mi), athletes can experience hyponatremia. A case study of two runners who collapsed after an ultramarathon race (160 km, or 100 mi) in 1983 revealed that their blood sodium concentrations had decreased from a normal value of 140 mmol/L to values of 123 and 118 mmol/L.¹⁹ One of the runners experienced a grand mal seizure; the other became disoriented and confused. Examining the runners' fluid intakes and estimating their sodium intakes during the run suggested that they had diluted their sodium contents by consuming too much fluid that contained too little sodium.

The ideal resolution to prevent hyponatremia would be to replace water at the exact rate at which it is being lost or to add sodium to the ingested fluid. The problem with the latter approach is that most sport drinks contain no more than 25 mmol/L of sodium and are apparently too weak to prevent sodium dilution alone, but very strong concentrations cannot be tolerated. Exercise hyponatremia appears to be the result of a fluid overload due to overconsumption, underreplacement of sodium losses, or both. Only a small number of cases have been reported. Thus, it is probably inappropriate to form conclusions from this information to design a fluid replacement regimen for people who must exercise for long periods in the heat.

Recommendations for Fluid Replacement Before, During, and After Exercise

Dehydration is a potential problem for athletes who train and compete for long periods of time, as well as in hot and humid environments. The American College of Sports Medicine (ACSM), the American

Dietetic Association, and the Dieticians of Canada have all published guidelines for adequate fluid intake before, during, and following exercise. Key recommendations include the following:

- Two hours before exercise, the athlete should consume 400 to 600 ml (14 to 22 oz.) of fluid to provide hydration and allow time for excretion of excess ingested water.
- During exercise, the athlete should drink enough fluid to keep fluid losses to less than 2% of body weight. Weight gain from overdrinking should be avoided.
- After exercise, the athlete should consume adequate fluids to fully replace sweat losses that occurred during exercise.
- Sport drinks containing carbohydrate concentrations of 4% to 8%, and sodium in concentrations between 0.5 and 0.7 g/L, are recommended during intense exercise events lasting longer than 1 h.
- Including sodium in drinks or eating high-sodium foods during the recovery period can help the rehydration process.²

In Review

- The need to replace lost body water is greater than the need to replace lost electrolytes, because sweat is very dilute.
- The thirst mechanism does not exactly match the body's hydration state, so often more fluid must be consumed than thirst dictates to maintain body weight.
- Fluid intake during prolonged exercise minimizes dehydration and optimizes cardiovascular and thermoregulatory function.
- In some rare cases, drinking too much fluid with too little sodium has led to hyponatremia (a low plasma concentration of sodium), which can cause confusion, disorientation, and even seizures, coma, and death.

Nutrition and Athletic Performance

Athletes place considerable demands on their bodies every day they train and compete. Optimal performance must include optimal nutrition. Too often, athletes spend considerable time and effort perfecting skills and attaining top physical condition while ignoring proper nutrition and sleep. Performance deterioration often can be traced to poor nutrition.

VIDEO 15.3 Presents Nancy Williams on reasons why some athletes might not eat enough.



Evidence-Based Nutrition Guidelines for Athletes

In 2016, experts from the American College of Sports Medicine, the American Dietetic Association, and the Dietitians of Canada collaborated to prepare a joint position statement on nutrition strategies for athletes.² This publication provides guidelines for the appropriate type, amount, and timing of foods, fluids, and dietary supplements. Based on an evidence-based analysis, their conclusions are summarized as follows:

1. Athletes need to consume energy that is adequate in amount and timing of intake during periods of high-intensity or long-duration training to maintain health and maximize training outcomes.
2. The primary goal of a training diet is to allow an athlete to stay healthy and injury-free while maximizing the functional and metabolic adaptations to training.
3. The optimal physique, including body size, shape, and composition (e.g., muscle mass and body fat levels), depends upon the sex, age, and heredity of the athlete, and may be sport and event specific. Where significant manipulation of body composition is required, it should ideally take place well before the competitive season to minimize the influence on event performance or reliance on rapid weight loss techniques.
4. Body carbohydrate stores provide an important fuel source for the brain and muscle during exercise and are manipulated by

exercise and dietary intake. Recommendations for carbohydrate intake typically range from 3 to 10 g/kg body weight per day (and up to 12 g/kg body weight per day for extreme and prolonged activities), but should be individualized to the athlete and situation.

5. Recommendations for protein intake typically range from 1.2 to 2.0 g/kg body weight per day, but have more recently been expressed in terms of the regular spacing of intakes of modest amounts of high-quality protein (0.3 g/kg body weight) after exercise and throughout the day. Such intakes can generally be met from food sources.
6. For most athletes, fat intakes associated with eating styles that accommodate dietary goals typically range from 20% to 35% of total energy intake. Consuming greater than or equal to 20% of energy intake from fat does not benefit performance, and extreme restriction of fat intake may limit the food range needed to meet overall health and performance goals.
7. Athletes should consume diets that provide at least the RDA or AI for all micronutrients.
8. A primary goal of competition nutrition is to address nutrition-related factors that may limit performance by causing fatigue and a deterioration in skill or concentration over the course of the event.
9. Foods and fluids consumed in the 1 to 4 h before an event should contribute to body carbohydrate stores, ensure appropriate hydration status, and maintain gastrointestinal comfort throughout the event.
10. Dehydration or hypohydration can increase the perception of effort and impair exercise performance; thus, appropriate fluid intake before, during, and after exercise is important for health and optimal performance. Individualized fluid plans should be developed.
11. An additional nutrition-related strategy for events of less than 60 min duration is to consume carbohydrate according to its potential to enhance performance.

12. Rapid restoration of performance between physiologically demanding training sessions or competitive events requires appropriate intake of fluids, electrolytes, energy, and carbohydrates to promote rehydration and restore muscle glycogen.
13. In general, vitamin and mineral supplements are unnecessary for athletes who consume a diet providing high energy availability from a variety of nutrient-dense foods. A multivitamin and mineral supplement may be appropriate in some cases when these conditions do not exist.
14. Athletes should be counseled regarding the appropriate use of ergogenic aids. These products (typically supplements) should be used only after careful evaluation for safety, efficacy, potency, and compliance with relevant antidoping codes and legal requirements.
15. Vegetarian athletes may be at risk for low intakes of energy, protein, fat, creatine, carnosine, n-3 fatty acids, and key micronutrients such as iron, calcium, riboflavin, zinc, and vitamin B₁₂.

Much of the scientific rationale for these recommendations is provided throughout this chapter and in the full position stand.² There are, however, special situations in which additional information is needed. We will now look at the precompetition meal and muscle glycogen replacement and loading.

Precompetition Meal

Although the meal ingested a few hours before competition might contribute little to muscle glycogen stores, which reflect long-term carbohydrate intake, it can ensure a normal blood glucose level and prevent hunger. This meal should contain only about 200 to 500 kcal and consist mostly of carbohydrate foods that are easily digested. Foods such as cereal, milk, juice, and toast are digested rather quickly and won't leave the athlete feeling full during competition. In general, this meal should be consumed at least 2 h before competition. The rates at which food is digested and nutrients are absorbed into the body are quite individual, so timing the precompetition meal might depend on prior experience. In one study

of endurance cyclists, a prolonged cycling exercise trial to exhaustion at 70% of the subject's $\dot{V}O_{2\max}$ was performed under two different conditions, with 14 days between trials: a breakfast containing 100 g of carbohydrate fed 3 h before exercise (fed) and no feeding before exercise (fasted). Subjects tested under the fed condition exercised 136 min before reaching exhaustion compared with 109 min in the fasted trial, indicating the importance of the precompetition meal.²⁹

A liquid precompetition meal might be less likely to result in nervous indigestion, nausea, vomiting, and abdominal cramps. Such feedings are commercially available and generally have been found useful both before and between events. Finding time for athletes to eat is often difficult when they must perform in multiple preliminary and final events. Under these circumstances, a liquid feeding that is low in fat and high in carbohydrate might be the only solution.

Muscle Glycogen Replacement and Loading

Earlier in this chapter, we established that different diets can markedly influence muscle glycogen stores and that endurance performance depends largely on these stores. The theory is that the greater the amount of glycogen stored, the better the potential endurance performance, because fatigue will be delayed. Thus, an athlete's goal is to begin an exercise bout or competition with as much stored glycogen as possible.

On the basis of muscle biopsy studies conducted in the mid-1960s, Åstrand⁴ proposed a plan to help runners store the maximum amount of glycogen. This process is known as glycogen or carbohydrate loading. According to Åstrand's regimen, athletes should begin to prepare for an aerobic endurance competition by completing an exhaustive training bout 7 days before the event. For the next 3 days, they should eat fat and protein almost exclusively to deprive the muscles of carbohydrate, which increases the activity of glycogen synthase, an enzyme responsible for glycogen synthesis and storage. Athletes should then eat a carbohydrate-rich diet for the remaining 3 days before the event. Because glycogen synthase activity is increased, increased carbohydrate intake results in greater muscle glycogen storage. Training intensity and volume during this 6-day period should be markedly reduced to prevent additional muscle glycogen depletion, thus maximizing liver and muscle glycogen

reserves. Originally, an additional intense training bout was performed 4 days before competition.

This regimen has been shown to elevate muscle glycogen stores to twice the normal level, but it is somewhat impractical for most highly trained competitors. During the 3 days of low carbohydrate intake, athletes generally find training difficult. They are also often irritable and unable to perform mental tasks, and they typically show signs of low blood sugar, such as muscle weakness and disorientation. In addition, the exhaustive depletion bouts of exercise performed 7 days before the competition have little training value and can impair glycogen storage rather than enhance it. This depletion exercise also exposes athletes to possible injury or overtraining.

Considering these limitations, many now propose that the depletion exercise and the low-carbohydrate aspects of Åstrand's regimen be eliminated. Instead, the athlete should simply reduce training intensity a week before competition and eat a normal, mixed diet containing 55% of the calories from carbohydrate until 3 days before the competition. For these days, training should be reduced to a daily warm-up of 10 to 15 min of activity and accompanied by a carbohydrate-rich diet. With this plan, as seen in [figure 15.16](#), glycogen will be elevated to nearly 200 mmol/kg of muscle, the same level attained with Åstrand's depletion regimen, and the athlete will be better rested for competition.

It is possible to increase carbohydrate stores rapidly after even a very short near-maximal-intensity bout of exercise. In a study of seven endurance athletes, scientists found that cycling for 150 s at 130% of $\dot{V}O_{2\text{peak}}$ followed by 30 s of all-out cycling and 24 h of high carbohydrate intake was sufficient to nearly double muscle glycogen stores in just one day.¹⁸

Diet is also important in preparing the liver for the demands of endurance exercise. Liver glycogen stores decrease rapidly when a person is deprived of carbohydrate for only 24 h, even when at rest. With only 1 h of strenuous exercise, liver glycogen decreases by 55%. Thus, hard training combined with a low-carbohydrate diet can empty the liver glycogen stores. A single carbohydrate meal, however, quickly restores liver glycogen to normal. Clearly, a carbohydrate-rich diet in the days preceding competition will

maximize the liver glycogen reserve and minimize the risk of hypoglycemia during the event.

Water is stored in the body at a rate of about 2.6 g of water with each gram of glycogen. Consequently, an increase or decrease in muscle and liver glycogen generally produces a change in body weight of 0.5 to 1.4 kg (1-3 lb). Some scientists have proposed monitoring changes in muscle and liver glycogen stores via recording the athlete's early morning weight immediately after rising—after emptying the bladder but before eating breakfast. A sudden decrease in weight might reflect a failure to replace glycogen, a deficit in body water, or both.

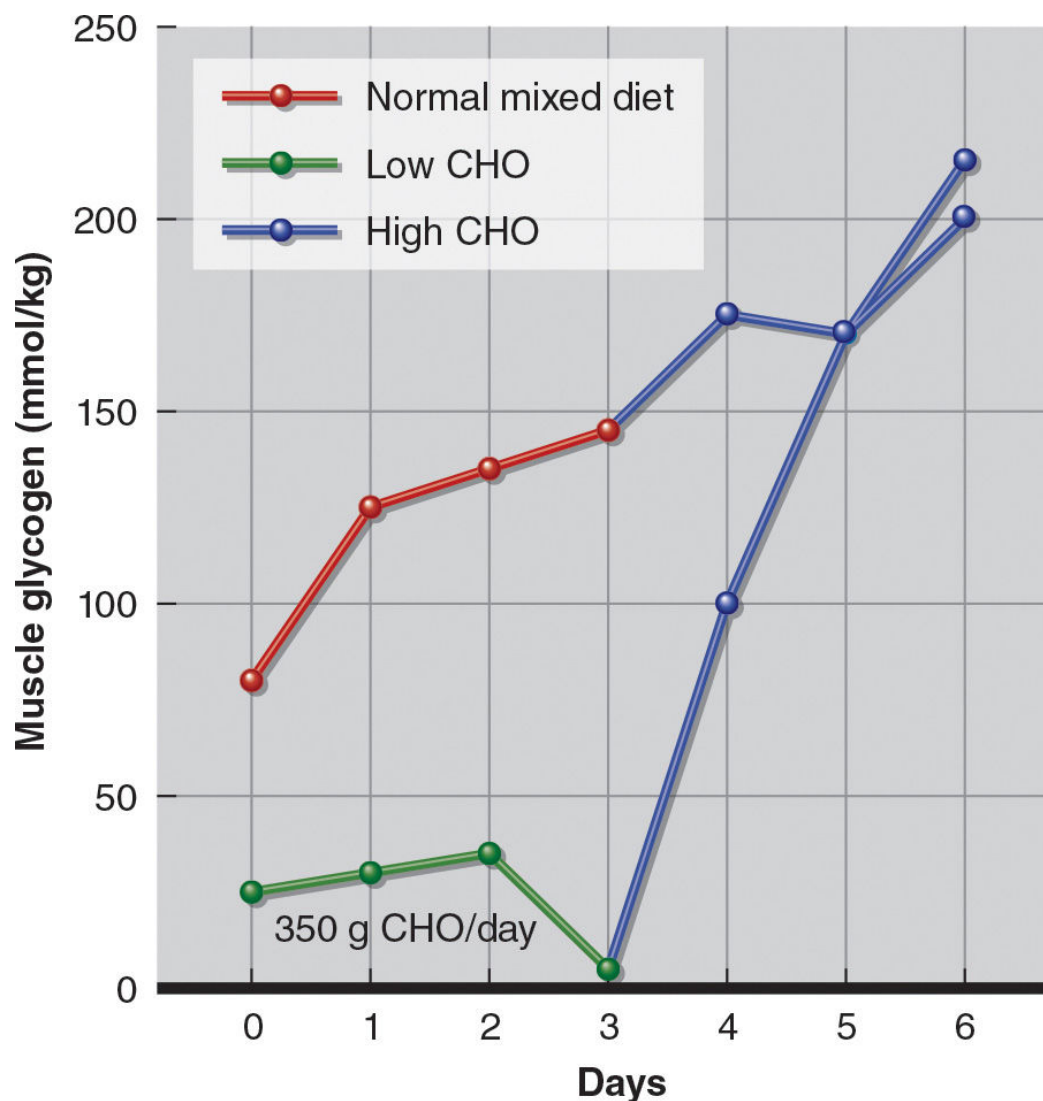


FIGURE 15.16 Two regimens for muscle glycogen loading. In one regimen, the subjects were depleted of muscle glycogen (day 0) and then ate a low-carbohydrate (CHO) diet for 3 days. They then switched

to a CHO-rich diet, which caused muscle glycogen to increase to about 200 mmol/kg. In the other dietary regimen, the subjects ate a normal, mixed diet and reduced their training volume for the first 3 days; they then changed to a high-CHO diet and a further reduction in training volume for 3 days, which also resulted in muscle glycogen of about 200 mmol/kg.

Data from Åstrand (1979).

Athletes who must train or compete in exhaustive events on successive days should replace muscle and liver glycogen stores as rapidly as possible. Although liver glycogen can be depleted totally after 2 h of exercise at 70% $\dot{V}O_{2\max}$, it is replenished within a few hours when a carbohydrate-rich meal is consumed. Muscle glycogen resynthesis, on the other hand, is a slower process, taking several days to return to normal after an exhaustive exercise bout such as the marathon (see [figure 15.17](#)). Studies in the late 1980s revealed that muscle glycogen resynthesis was most rapid when individuals were fed at least 50 g (about 0.7 g/kg body weight) of glucose every 2 h after the exercise.²² Feeding subjects more than this amount did not appear to accelerate the replacement of muscle glycogen. During the first 2 h after exercise, the rate of muscle glycogen resynthesis is much higher than later in recovery, as discussed earlier in this chapter. Thus, an athlete recovering from an exhaustive endurance event should ingest sufficient carbohydrate as soon after exercise as is practical. Adding protein and amino acids to the carbohydrate ingested during the recovery period enhances muscle glycogen synthesis above that achieved with carbohydrate alone.

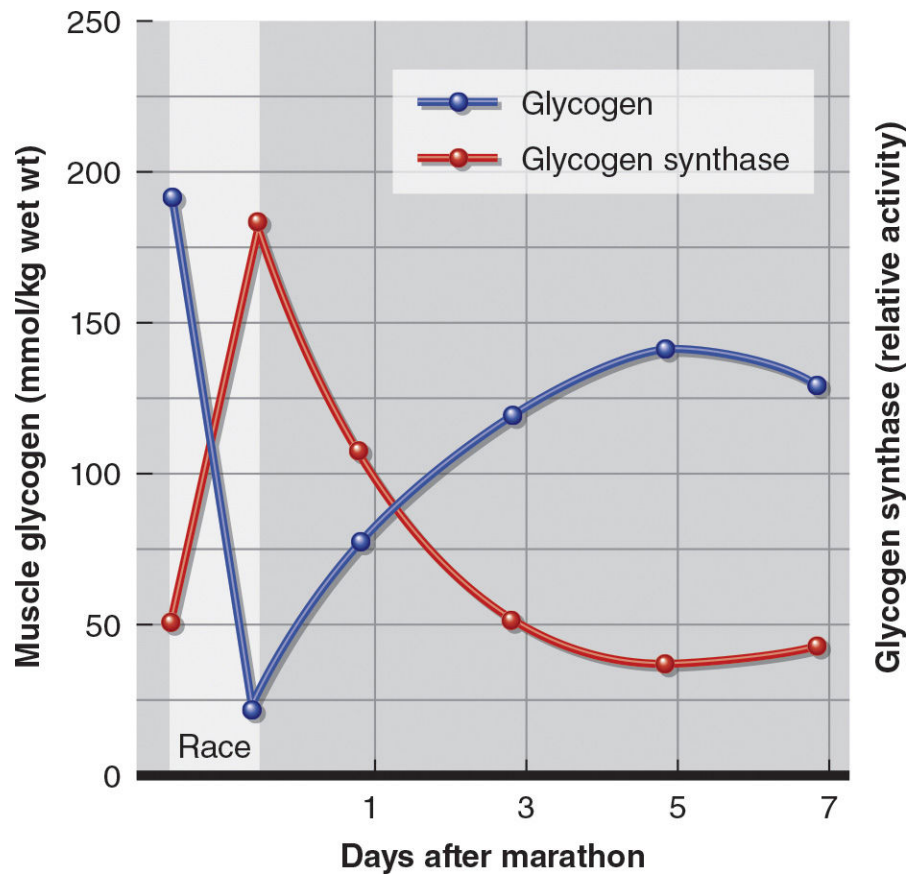


FIGURE 15.17 Muscle glycogen resynthesis is a slow process, requiring several days to restore normal muscle glycogen storage following exhaustive exercise. Note that when muscle glycogen decreases with hard exercise (race), muscle glycogen synthase is markedly elevated. This triggers the muscle to store glycogen when carbohydrates are eaten, returning glycogen synthase to the baseline level.

Sport Drinks

We mentioned earlier that ingesting carbohydrate before, during, and following exercise benefits performance by ensuring adequate fuel for energy production during exercise and replenishing glycogen stores following exercise. While selecting a wise diet can provide for most of the athlete's nutrition needs, nutritional supplements can also be of great value. In addition, adequate fluid intake is necessary for preexercise hydration, hydration during exercise, and rehydration following exercise. Sport drinks are uniquely designed to meet both the energy and fluid needs of the athlete. Performance benefits from these drinks have been clearly documented, not only in endurance activities but also in burst activities as well (e.g., soccer and basketball).^{10,17}

Sport drinks differ from one another in a number of ways besides in taste. Of major concern, however, is the rate at which energy and water are delivered. Energy delivery is primarily determined by the concentration of the carbohydrates in the drink, and fluid replacement is influenced by both taste and the sodium concentration of the drink.

In Review

- Some athletes have adopted vegetarian diets and appear to perform well. However, careful consideration must be given to protein sources and to consuming adequate levels of iron, zinc, calcium, and several vitamins.
- The precompetition meal should be eaten no less than 2 h before competition, and it should be low in fat, high in carbohydrate, and easily digestible. A liquid precompetition meal low in fat and high in carbohydrate has advantages.
- Carbohydrate loading increases muscle glycogen content, which in turn increases endurance performance.
- After endurance competition or training, it is important to consume substantial carbohydrate to replace the glycogen used during activity. Replacing glycogen during the first few hours after training or competing is optimal because glycogen synthase levels are at their peak.

Energy Delivery: The Carbohydrate Concentration

A major concern is how rapidly the drink leaves the stomach, or the rate of **gastric emptying**. In general, carbohydrate solutions empty more slowly from the stomach than either water or a weak sodium chloride (salt) solution. Research suggests that a solution's caloric content, a reflection of its concentration, might be a major determinant of how quickly it empties from the stomach and is absorbed from the intestine. Since carbohydrate solutions remain in the stomach longer than either water or weak solutions, increasing the glucose concentration of a sport drink significantly reduces the gastric emptying rate. For example, 400 ml (14 oz) of a weak glucose solution (139 mmol/L) is almost completely emptied from the stomach in 20 min, but emptying a similar volume of a strong glucose solution (834 mmol/L) can require nearly 2 h.¹⁴ However, when even a small amount of a strong glucose drink leaves the stomach, it can contain more sugar than a larger amount of a weaker solution simply because of its higher concentration. But, if an athlete is trying to

prevent dehydration, this would deliver less water and thus be counterproductive.

What is the optimal CHO ingestion rate, that is, what dose of CHO is most beneficial to athletic performance? The ACSM and the National Athletic Trainers' Association both recommend CHO ingestion at a rate of 30 to 60 g/h for endurance athletes, but both lower and higher CHO doses have also been shown to improve performance. The optimal concentration would depend on several factors, including emptying from the stomach, absorption into the blood from the intestines, and the rate at which muscle can use the CHO (about 60-90 g/h depending on the type of CHO ingested). Concentrations of CHO that are too high can cause stomach discomfort and slow absorption of both CHO and fluid from the intestines.

A 2013 study³⁰ investigated the relation between CHO concentration in sport drinks and cycling time-trial performance. Fifty-one young cyclists and triathletes each completed four trials. After a 2 h ride at about 70% $\dot{V}O_{2peak}$, subjects performed a computer-simulated 20 km time trial, which they were asked to complete as quickly as possible. Twelve different beverages were consumed (1 L/h) during the 2 h ride in a double-blind manner (neither the person giving the athlete the beverage nor the athlete knew which beverage was which). The beverages provided subjects with 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, or 120 g of mixed CHO (1:1:1 glucose-fructose-maltodextrin) per hour or a noncaloric placebo. Subsequent time-trial performance improved incrementally with CHO concentrations up to 78 g/h, with diminishing performance enhancement at CHO levels greater than 78 g/h. That is, the performance-enhancing qualities of the sport drinks started to plateau at CHO intakes above 78 g/h, suggesting that in competitive race settings, athletes should strive to consume a mixed-CHO sport drink at a CHO ingestion rate of 68 to 88 g/h.

Most sport drinks on the market contain about 6 to 8 g of carbohydrate per 100 ml (3.5 oz) of fluid (6% to 8%). The carbohydrate source is generally glucose, glucose polymers, or a combination of glucose and glucose polymers, although fructose or sucrose has also been used.²² Research studies have confirmed

enhanced endurance performance with use of solutions in this range of concentration and with these sources of carbohydrate when compared to water.² Carbohydrate solutions above ~6% slow gastric emptying and limit the immediate availability of fluid. However, they can provide a greater amount of carbohydrate in a given period of time to meet the increased energy needs.^{2,23}

Rehydration with Sport Drinks: The Sodium Concentration

Just adding fluid to the body during exercise lessens the risk of serious dehydration. But research indicates that adding glucose and sodium to sport drinks, aside from supplying an energy source, stimulates both water and sodium absorption. Sodium increases both thirst and the palatability of the drink. Recall that when sodium is retained, this causes more water to be retained. For rehydration purposes, both during and following exercise, the sodium concentration should range between 20 and 60 mmol/L.²³ There is an important loss of sodium from the body with sweating. With high rates of sweating and large volumes of water intake, this can lead to critical reductions in the sodium concentration of the blood and possibly to hyponatremia, as discussed earlier in this chapter.

Palatability

Athletes will not drink solutions that taste bad. Unfortunately, we all have different taste preferences. To further confound the issue, what tastes good before and after a long, hot bout of exercise will not necessarily taste good during the event. Studies of taste preferences of runners and cyclists during 60 min of exercise showed that most chose a drink with a light flavor and no strong aftertaste. But, will athletes drink more if given a sport drink as compared to water? In one study, runners ran on a treadmill for 90 min and then recovered while seated for an additional 90 min. Both exercise and recovery conditions were controlled in an environmental chamber at a temperature of 32 °C (86 °F), 50% humidity. Three trials were conducted, two with two different sport drinks (6% and 8% carbohydrate) and one with water. Subjects were encouraged to drink throughout each trial. The volume consumed during exercise was similar for all three drinks, but during recovery, the runners drank about 55% more of each of the two sport drinks than water.³⁷

In Review

- Sport drinks have been shown to reduce the risk of dehydration and provide an important source of energy. They also can improve the performance of the athlete in both endurance and sudden burst activities such as soccer and basketball.
 - To maximize both CHO and fluid intake, the carbohydrate concentration of a sport drink generally should not exceed 6% to 8%.²
 - In competitive race settings, athletes should strive to consume a mixed-CHO sport drink at a CHO ingestion rate of 68 to 88 g/h.
 - The inclusion of sodium in a sport drink facilitates the intake and storage of water.
 - Taste is an important factor when one is considering a sport drink. Most athletes prefer a light flavor without a strong aftertaste. Each athlete should select the drink that tastes best, provided that the nutritional ingredients are the same.
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IN CLOSING

In this chapter, we examined the body composition and nutrition needs of the athlete, considering the importance of optimizing body composition for peak performance and eating wisely to enhance athletic performance. We discovered the importance of each of the six nutrient categories and how they can be adjusted to meet the athlete's training and performance needs. We looked at the precompetition meal, ways to effectively replenish and load muscle glycogen stores, and the effectiveness of commercial sport drinks. Now that we have a better understanding of the importance of an appropriate weight and a balanced diet, we turn our attention to another aspect of the athlete's quest for success. In the next chapter, we evaluate substances that have been proposed to enhance athletic performance—ergogenic aids.

KEY TERMS

air plethysmography
bioelectric impedance
body composition
body density
carbohydrate loading
dehydration
densitometry
dual-energy X-ray absorptiometry (DEXA)
electrolyte
essential amino acids
extracellular fluid
fat
fat-free mass
fat mass
free radicals
gastric emptying
glycogen loading
hydrostatic weighing
hyponatremia
intracellular fluid
macrominerals
microminerals
nonessential amino acids
osmolarity
protein

relative body fat
skinfold fat thickness
thirst mechanism
trace elements
vitamin

STUDY QUESTIONS

1. Differentiate between body size and body composition. What tissues of the body constitute the fat-free mass?
2. What is densitometry? How is it used to assess the body composition of the athlete? What is the major weakness of densitometry with respect to its accuracy?
3. What are several field techniques for estimating body composition? What are their strengths and weaknesses?
4. What is the relationship of relative leanness and fatness to performance in sport?
5. What guidelines should be used to determine the athlete's goal weight?
6. What are the six categories of nutrients?
7. What role does dietary carbohydrate play in endurance performance? How about fat? Protein?
8. When consuming carbohydrates during competition, what is the optimal dose that athletes should strive to consume?
9. What is an appropriate protein allowance for a normally active adult man? For a woman? Discuss the value of using protein supplements to enhance performance in strength and endurance events.
10. Should the athlete supplement vitamins and minerals?
11. How does dehydration affect exercise performance? What effect does dehydration have on exercise heart rate and body temperature?
12. Describe the recommended precompetition meal.
13. Describe the method used to maximize muscle glycogen storage (glycogen loading).
14. Discuss the value of consuming carbohydrate during and after endurance exercise. What are the potential benefits of sport drinks?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

